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Semi-Empirical Prediction of Aircraft Low-Speed Aerodynamic Characteristics

This paper lays out a comprehensive methodology for computing a low-speed, high-lift polar, without requiring additional details about the aircraft design beyond what is typically available at the conceptual design stage. Introducing low-order, physics-based aerodynamic analyses allows the methodology to be more applicable to unconventional aircraft concepts than traditional, fully-empirical methods. The methodology uses empirical relationships for flap lift effectiveness, chord extension, drag-coefficient increment and maximum lift coefficient of various types of flap systems as a function of flap deflection, and combines these increments with the characteristics of the unflapped airfoils. Once the aerodynamic characteristics of the flapped sections are known, a vortex-lattice analysis calculates the three-dimensional lift, drag and moment coefficients of the whole aircraft configuration. This paper details the results of two validation cases: a supercritical airfoil model with several types of flaps; and a 12-foot, full-span aircraft model with slats and double-slotted flaps.

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Authors
Olson, Erik D.
(NASA Langley Research Center Hampton, VA, United States)

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